

Reading Plan — The Realization-Theorem Technique (the σ -essential fourth-cell route)

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Why this plan exists

This is a *technique-acquisition* plan, companion to the broader `sigma_essential_skill_plan` (which sequences descriptive set theory and large cardinals). It is narrower and pointed at one route.

The σ -essential import sweep closed every *existing* object one could borrow as the carrier’s skeleton: each dies either Boolean-ambient (rescued by a non-Dirac σ -state) or operator-algebraic (non-concrete by Kochen–Specker). The one surviving *lead* — not a solution, a place to learn — is the family of **realization theorems**. Per the prior-art note (`czech_school_prior_art_sigma_essential.md` §POSITIVE), these build *non-simplex* (hence contextual, non-segregated) state spaces on orthomodular lattices by *pasting Boolean blocks* — a combinatorial, non-operator-algebraic construction, so the orthocomplement is block-inherited and the route dodges the polarity/form horn that kills matroids and geometric lattices.

The one crux — read every source against it. The realization theorems in the literature are *finite-pasting* constructions and realize the *full (finitely additive)* state space. They fail the σ -essential target on a single **conjunction gap**:

*Why must the pasting be finite, and can an infinite pasting realize a non-simplex state space on an OML that is **simultaneously** concrete and σ -complete?*

No located paper does this (prior-art note, §POSITIVE). That conjunction — concrete + σ -complete + non-simplex-surviving-restriction-to- σ -states — is exactly the open cell. Do not read these sources for “realization theory” in general; read each only for how it bears on the conjunction gap.

Citation caveat. The realization papers below are **not yet in the repo bib**; venue/year/page are reconstructed from the prior-art note and are marked [cite unverified]. Verify against the primary source before relying on any detail. Claims about paper *content* here are the prior-art note’s summary, not a reading of the papers themselves.

1 Foundations: how block-pasting builds an OML

Kalmbach, *Orthomodular Lattices* (or Pták–Pulmannová, *Orthomodular Structures as Quantum Logics* [1] for the concrete-logic side). The pasting construction and horizontal sums.

- State the pasting of a family of Boolean blocks precisely: how blocks overlap, what the resulting order and orthocomplement are. *Confirm* the orthocomplement is inherited block-wise, not derived from any form — this is why the route dodges the scout’s polarity horn.

- Where does the loop / commutativity condition force the paste to be an OMP rather than a lattice? (This is one place “infinite non-segregated pasting” can silently fail before σ even enters.)
- Contrast a *segregated* paste (horizontal sum, contextuality on a central factor — $\prod_n MO_2$, MO_κ) with a non-segregated one. What structural feature makes the incompatibility non-central? The target needs the latter.

2 The realization theorems (read for: where finiteness enters)

Navara–Rogalewicz, “The pasting constructions for orthomodular posets,” *Demonstratio Math.* (1988) [cite unverified]. The base realization: every (compact convex) set arises as a state space of a pasted OMP.

- Read for the *exact step* where finiteness of the pasting is used to realize a non-simplex state space. Which part of the argument breaks if the family of blocks — or an individual block — is infinite?
- The realized object is the *full, finitely additive* state space. Is there any control on σ -additive states? (Prior-art note: no — this is the unaddressed point.)

Harding–Navara, “Subalgebras of orthomodular lattices,” *Order* 17 (2000) [cite unverified]. The strengthened realization: prescribed center and automorphism group.

- “Prescribed center” is the lever for *non-segregation*: a trivial (or small) center forces contextuality off any central Boolean factor. Read for how the center is controlled, and whether that control survives an infinite paste.
- Same finiteness question as above: locate the finite-pasting dependency.

3 The concreteness half (read for: what buys set-representability)

Mayet–Navara (1995) [cite unverified]. Concreteness for pasted OMLs holds only for special forms (prior-art note names $K(L)$).

- What is the special form that yields a set-representable (concrete) paste, and is it compatible with the *infinite* pasting the σ -side needs? Concrete + σ together is the whole gap; this is the paper on the concreteness half.
- Concreteness = enough two-valued states to separate points ($a \mapsto \{s : s(a) = 1\}$ is an embedding). Does the $K(L)$ form preserve a separating supply of *two-valued* states, or only states in general?

4 The σ -boundary (read for: what survives restriction to σ -states)

Navara–Rüttimann, “A characterization of σ -state spaces of orthomodular lattices,” *Expo. Math.* 9 (1991) 275–284 [cite unverified; NOT in repo bib — the existing Rüttimann1994 key is a different Rüttimann paper (“Weakly purely finitely additive measures,” *Canad. J. Math.* 46), do not conflate].

- The theorem places $S_\sigma(P)$ as a *semi-exposed face* of the full state space $S(P)$ (per Pták’s zbMATH review). Read for: does a *non-simplex* full state space, restricted to this face, stay non-simplex? A face of a non-simplex set need not be a simplex — so this is the gate the realized non-simplex must pass to remain contextual for σ -additive states.

- This is the precise juncture where the finite-pasting realization would have to be shown to survive σ -restriction. It is the mathematical heart of the push.

5 The impossibility twin (one entry — do not over-invest)

The same σ -state-space characterization (Navara–Rüttimann 1991) is also the tool for the *negative* attempt: if the semi-exposed face is forced to be a simplex with dispersion-free vertices on every concrete σ -complete OML, the cell is empty. The prior-art note records that the verbatim theorem does *not* force this — so the emptiness proof, if it exists, needs more. Read the characterization once for both directions; the new-world and impossibility branches stay indistinguishable until one is worked.

The order

Kalmbach/Pták–Pulmannová (pasting mechanics) $\rightarrow$ Navara–Rogalewicz then Harding–Navara (locate the finiteness dependency) $\rightarrow$ Mayet–Navara (concreteness form) $\rightarrow$ Navara–Rüttimann (the σ -restriction gate). The push, if it works, is a single new theorem: an *infinite*, concrete, σ -complete paste whose non-simplex state space survives restriction to σ -additive states. Everything above is read to find where that theorem must do its work, and where the finite-pasting literature stops.

References

- [1] Pavel Pták and Sylvia Pulmannová. *Orthomodular Structures as Quantum Logics*, volume 44 of *Fundamental Theories of Physics*. Kluwer, Dordrecht, 1991.